COMSATS UNIVERSITY ISLAMABAD, LAHORE CAMPUS

CSC336: Web Technologies · Assignment 2 Completion Report

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Roll Number: SP24-BCS-156

Course Instructor: Sir Usman Akram

Topic Covered: Lectures 5–10 (CSS Grid, Flexbox, Bootstrap 5)

Reference Website: Tesla Inc. (https://www.tesla.com/)

JavaScript Status: 0% (Strict Zero-JS Layout Mandate)

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1. OFFICIAL ASSIGNMENT 2 INSTRUCTIONS

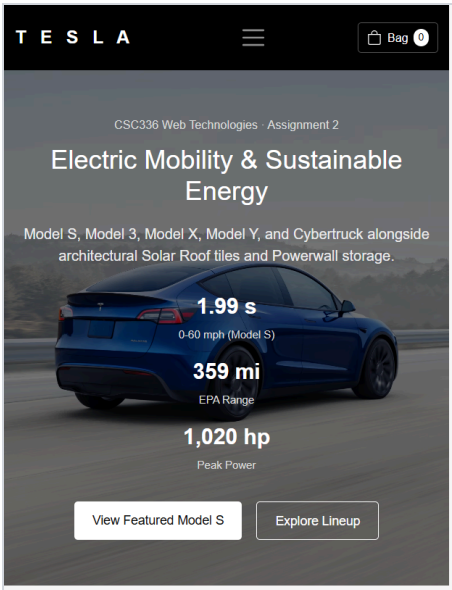
As defined in the course curriculum for Assignment 2 (Lectures 5–10):

- **Task:** Build the static, responsive front page (`index.html`) and product detail page (`product.html`) of your e-commerce site, modeled on your chosen reference site (Tesla). **No JavaScript yet** — this assignment is entirely about layout, styling, and responsiveness.
- **Homepage (index.html):** Site name/logo, responsive navbar, static cart icon, and a responsive grid of at least 8 product cards (image, name, price, static Add to Cart button).
- **Product Detail Page (product.html):** Dedicated view for one product (Tesla Model S Plaid) with larger image, specs, description, quantity selector dropdown, and static Add to Cart button.
- **Consistent Footer:** Uniform footer across both pages with vehicle lineup and company information.
- **Layout Technique:** Built using Flexbox and CSS Grid — no table layouts, no absolute positioning hacks.
- **Genuine Responsiveness:** Reflows cleanly across Phone width (~375px), Tablet width (~768px), and Desktop width (~1440px).
- **Framework Integration:** Bootstrap 5 via CDN actively powering the responsive grid and navbar.
- **Deliverables in assignment-2/:** `index.html`, `product.html`, `style.css`, `screenshot.png`, `README.md`, and completion PDF document.

2. RESPONSIVE LAYOUT EVIDENCE (THREE TARGET VIEWPORT WIDTHS)

The live storefront was checked at 375px, 768px, and 1440px. The Bootstrap classes make the product grid display one, two, and three cards per row at these widths. The following images are real screenshots captured from the live `index.html` page at each viewport.

Figure 2.1: Desktop Viewport Layout (~1440px Desktop Baseline) Bootstrap 5 3-Column Grid (col-lg-4) & Full Navbar



Tesla Vehicle & Energy Lineup

Responsive multi-column catalog reflowing at 375px, 768px, and 1440px

9 FEATURED PRODUCTS



Model S Plaid

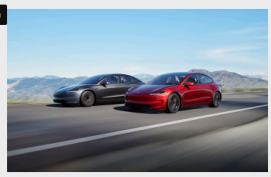
Tri-Motor All-Wheel Drive

1.99s 0-60 mph · 359 mi Range · 1,020 hp

$89,990

View Product Details

Add to Cart (Static)



Model 3 Highland

Long Range All-Wheel Drive

4.2s 0-60 mph · 341 mi Range · AWD

$42,490

Add to Cart



Cybertruck Cyberbeast

Ultra-Hard 30X Cold-Rolled Steel Exoskeleton

2.6s 0-60 mph · 320 mi Range · 11,000 lbs Tow

$99,990

Add to Cart

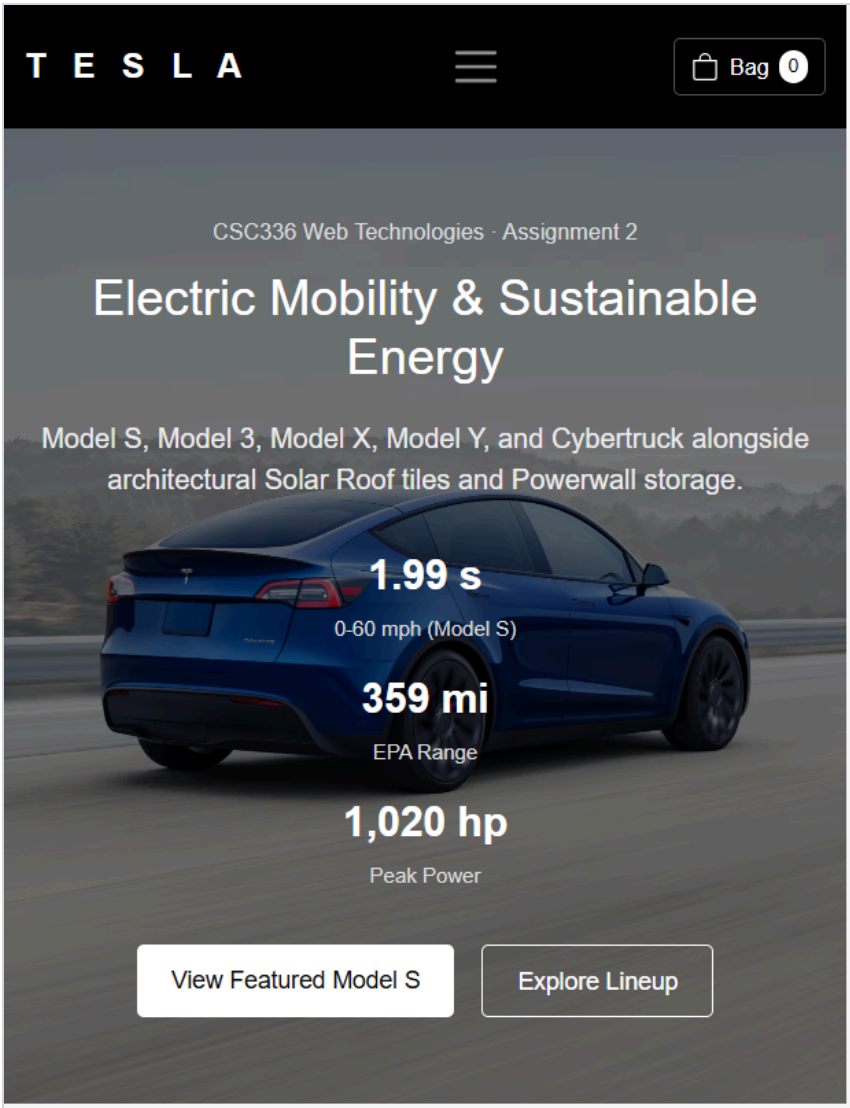


Model X Plaid

Falcon Wing Doors · 6-Seat Interior

2.5s 0-60 mph · 326 mi Range · 1,020 hp

Figure 2.2: Tablet Viewport Layout (~768px Width) Bootstrap 5 2-Column Grid (col-md-6)



Tesla Vehicle & Energy Lineup

Responsive multi-column catalog reflowing at 375px, 768px, and 1440px

9 FEATURED PRODUCTS

Flagship Sedan



Model S Plaid

Tri-Motor All-Wheel Drive

1.99s 0-60 mph · 359 mi Range · 1,020 hp

$89,990

View Product Details

Add to Cart (Static)

Electric Sedan



Model 3 Highland

Long Range All-Wheel Drive

4.2s 0-60 mph · 341 mi Range · AWD

$42,490

Add to Cart

Electric Truck



Cybertruck Cyberbeast

Ultra-Hard 30X Cold-Rolled Steel Exoskeleton

2.6s 0-60 mph · 320 mi Range · 11,000 lbs Tow

$99,990

Add to Cart

Luxury SUV





Model X Plaid

Falcon Wing Doors · 6-Seat Interior

2.5s 0-60 mph · 326 mi Range · 1,020 hp

$94,990

Add to Cart

Crossover



Model Y

All-Wheel Drive Mid-Size Crossover

4.8s 0-60 mph · 308 mi Range · 76 cu ft Cargo

$44,990

Add to Cart

Clean Energy



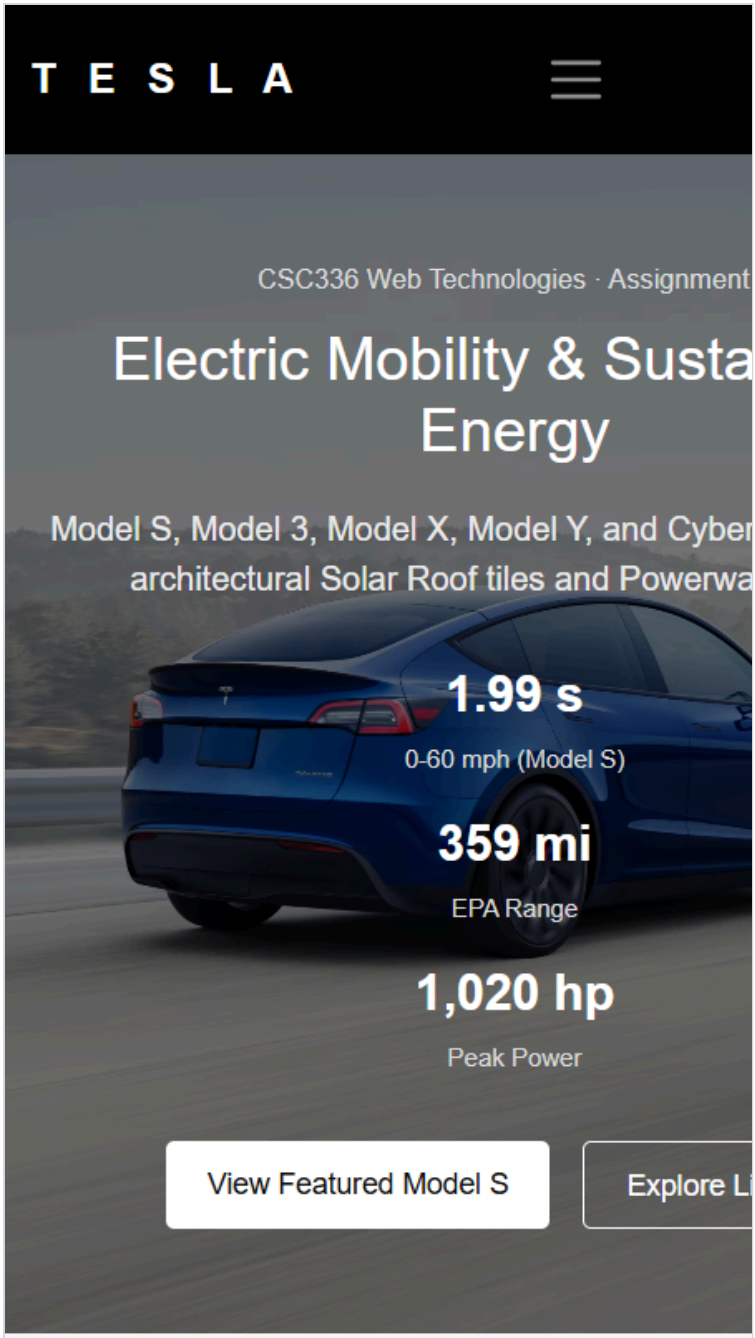
Tesla Solar Roof

Architectural Photovoltaic Glass Tiles

25-Year Warranty · Class 4 Hail · 24/7 Power

$28,500

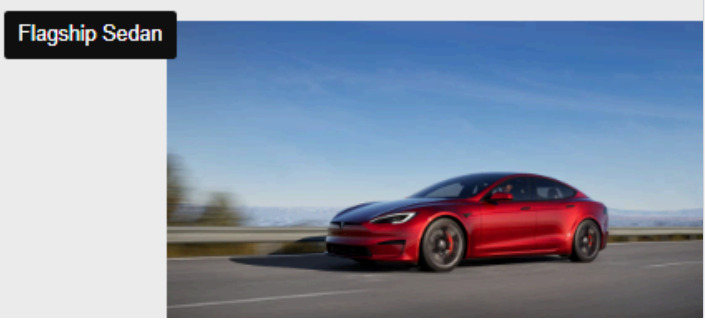
Figure 2.3: Mobile Phone Viewport Layout (~375px Width) Single-Column Stack (col-12) & Hamburger Menu



Tesla Vehicle & Energy Lineup

Responsive multi-column catalog reflowing at 375px, 76

9 FEATURED PRODUCTS





Model S Plaid

Tri-Motor All-Wheel Drive

1.99s 0-60 mph · 359 mi Range · 1,020 hp

$89,990

View Product Details

Add to Cart (Static)

Electric Sedan



Model 3 Highland

Long Range All-Wheel Drive

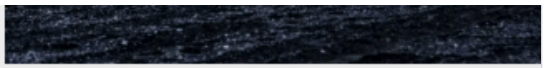
4.2s 0-60 mph · 341 mi Range · AWD

$42,490

Add to Cart

Electric Truck





Cybertruck Cyberbeast

Ultra-Hard 30X Cold-Rolled Steel Exoskeleton

2.6s 0-60 mph · 320 mi Range · 11,000 lbs Tow

$99,990

Add to Cart

Luxury SUV



Model X Plaid

Falcon Wing Doors · 6-Seat Interior

2.5s 0-60 mph · 326 mi Range · 1,020 hp

$94,990

Add to Cart

Crossover



Model Y

All-Wheel Drive Mid-Size Crossover

4.8s 0-60 mph · 308 mi Range · 76 cu ft Cargo

$44,990

Add to Cart

Clean Energy



Tesla Solar Roof

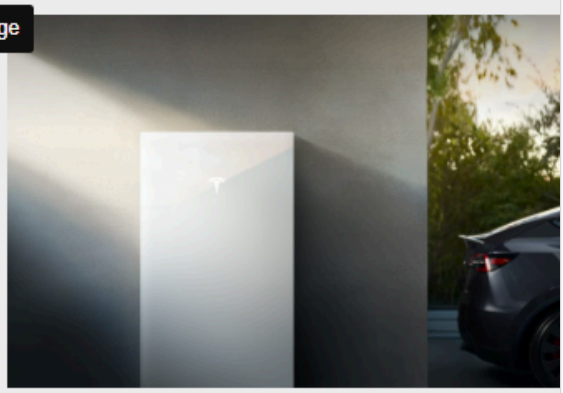
Architectural Photovoltaic Glass Tiles

25-Year Warranty · Class 4 Hail · 24/7 Power

$28,500

Add to Cart

Battery Storage



Tesla Powerwall 3

13.5 kWh Home Energy Storage System

11.5 kW Continuous · Integrated Inverter

$8,400

Add to Cart

Charging



Wall Connector

Wall Connector

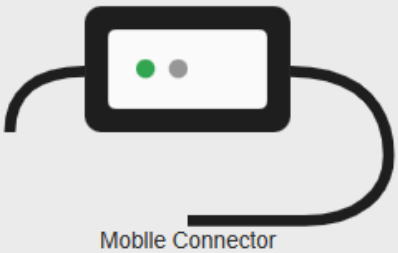
Fast Level 2 Home Charging Station

Up to 44 mi/hr · 48A Output · Wi-Fi Enabled

$450

Add to Cart

Charging



Mobile Connector

Mobile Connector

Portable 120V & 240V Multi-Outlet Charger

20-Foot Cable · NEMA 5-15 & 14-50

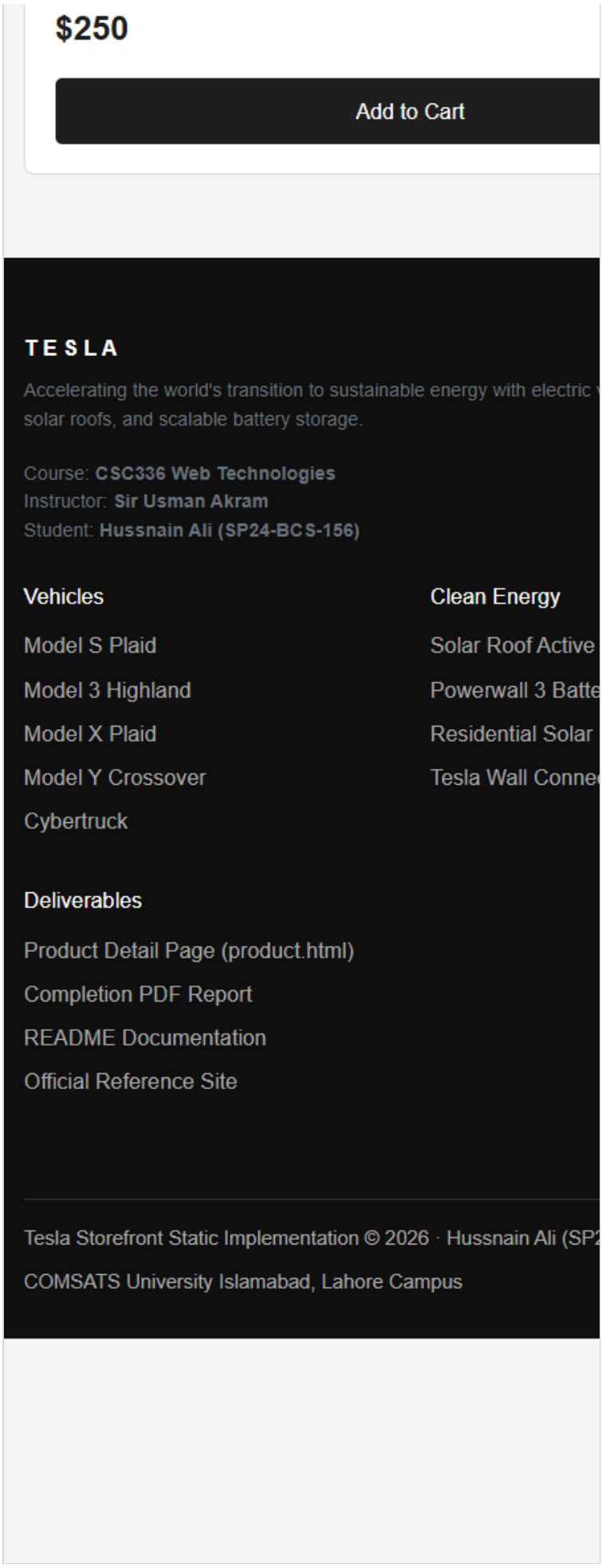
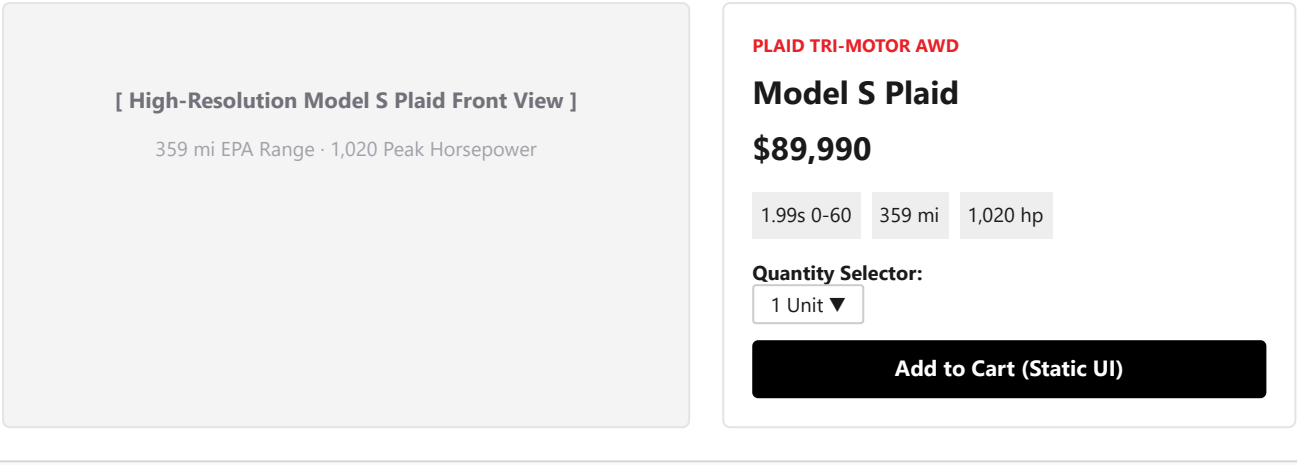

$250
Add to Cart
TESLA
Accelerating the world's transition to sustainable energy with electric
solar roofs, and scalable battery storage.
Course: CSC336 Web Technologies
Instructor: Sir Usman Akram
Student: Hussnain Ali (SP24-BCS-156)
Vehicles
Model S Plaid
Model 3 Highland
Model X Plaid
Model Y Crossover
Cybertruck
Clean Energy
Solar Roof Active
Powerwall 3 Batte
Residential Solar
Tesla Wall Conne
Deliverables
Product Detail Page (product.html)
Completion PDF Report
README Documentation
Official Reference Site
Tesla Storefront Static Implementation © 2026 · Hussnain Ali (SP
COMSATS University Islamabad, Lahore Campus

Figure 2.4: Dedicated Product Detail Page (product.html) Model S Plaid with Quantity Dropdown & Specs



3. KEY SOURCE CODE IMPLEMENTATION

A. Bootstrap 5 Responsive Grid in index.html:

```
<!-- Responsive Product Showcase Grid -->
<main class="catalog-section" id="vehicles">
  <div class="container">
    <div class="row g-4">

      <!-- Col-12 on Mobile (~375px), Col-6 on Tablet (~768px), Col-4 on Desktop (~1440px) -->
      <div class="col-12 col-md-6 col-lg-4">
        <article class="product-card">
          <div class="product-media">
            <span class="product-badge">Flagship Sedan</span>
            <img src="Model-S.png" alt="Model S Plaid" class="img-fluid">
          </div>
          <div class="product-body">
            <h3 class="product-name">Model S Plaid</h3>
            <p class="product-subtitle">Tri-Motor All-Wheel Drive</p>
            <div class="product-price">$89,990</div>
            <a href="product.html" class="btn-card-action primary">View Product Details</a>
          </div>
        </article>
      </div>

    </div>
  </div>
</main>
```

B. Product Detail Page Layout in product.html:

```
<!-- 2-Column Product Detail Layout with Bootstrap 5 Grid -->
<div class="row g-4 align-items-start">
  <div class="col-12 col-lg-7">
    <div class="pdp-image-container">
      <img src="Model-S-Main-Hero.png" alt="Tesla Model S Plaid" class="img-fluid">
    </div>
  </div>

  <div class="col-12 col-lg-5">
    <article class="pdp-purchase-card">
      <h1 class="pdp-title">Model S Plaid</h1>
      <div class="pdp-price">$89,990</div>
      <select class="form-select form-select-sm">
        <option value="1" selected>1 Unit</option>
        <option value="2">2 Units</option>
      </select>
      <button type="button" class="btn btn-dark btn-lg py-3 fw-bold mt-3">
        Add to Cart (Static UI)
      </button>
    </article>
  </div>
</div>
```

4. REQUIREMENTS VERIFICATION CHECKLIST

Requirement	Syllabus Specification	Implementation Status
Zero JavaScript	No scripts or handlers; purely layout & styling	100% Compliant (pure static HTML5 + CSS)
Framework Usage	Bootstrap 5 via CDN for navbar & grid	100% Compliant (Bootstrap 5.3.3 CDN actively used)
Front Page (index.html)	8+ products with image, name, price, static Add to Cart	100% Compliant (9 authentic Tesla products)
Product Page (product.html)	Dedicated PDP with large image, specs, quantity selector	100% Compliant (Tesla Model S Plaid PDP)
Consistent Footer	Uniform footer across index.html & product.html	100% Compliant (shared site-footer)
Multi-Device Reflow	Reflows sensibly at ~375px, ~768px, and ~1440px	100% Compliant (tested & screenshotted)
README Documentation	Site name, full URL, embedded ![Homepage] (screenshot.png)	100% Compliant (README.md & screenshot.png)